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(1) Research Program/Project/Study Title: Assessing the Acceptability of Pineapple Concentrate through Incorporation of Mature Coconut Water: A Sensory, Physico-Chemical, and Consumer Perception Study	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ BSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Roxas City
Province:		Capiz
Region:		6
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)

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(7) Sector/Commodity	(8) Discipline/Field of Specialization

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(9) Rationale

Food waste is a critical global issue, with implications spanning economic, environmental, and social domains. Throughout the food supply chain, from production to consumption, significant amounts of food are lost or discarded, exacerbating inefficiencies and sustainability challenges. To address this issue, innovative approaches are imperative, leveraging underutilized resources and promoting sustainable practices.

In the Philippines, the coconut industry stands as a cornerstone of the economy, producing valuable commodities such as virgin coconut oil, coconut milk, and dried coconut. However, a substantial byproduct of this process is mature coconut water, often overlooked and discarded as waste due to prevalent processing methods. This overabundance of mature coconut water poses environmental concerns, necessitating strategies for its utilization to mitigate waste and maximize resource efficiency.

Other agricultural products, like pineapple, offer opportunities for value addition and waste reduction. Pineapple, renowned for its nutrient density and health benefits, presents versatile applications, from dietary consumption to industrial uses such as food coloring. Moreover, seafood, particularly seaweed, holds immense potential for enhancing nutritional value and functional properties in food and beverage products. Seaweed is a rich source of essential vitamins, minerals, and bioactive compounds, offering numerous health benefits ranging from improved digestion to enhanced immune function. Furthermore, seaweed cultivation has minimal environmental impact, making it a sustainable option for ingredient sourcing.

In light of these considerations, this study explores the utilization of matured coconut water and pineapple concentrate, enriched with seaweed extract, in the development of functional drinks. By integrating these underutilized resources and harnessing the nutritional benefits of seaweed, this research aims to reduce food waste, promote sustainable resource utilization, and enhance the nutritional quality of functional beverages. This research aims to offer practical solutions for addressing food waste, promoting sustainable agriculture, and advancing the nutritional value of functional beverages with a particular focus on the utilization of seaweed, mature coconut water and pineapple with its health-promoting properties and environmental sustainability.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

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(10) Objectives/Statement of the Problem

The primary aim of this investigation is to scientifically evaluate the acceptability of incorporating mature coconut water into the production of pineapple concentrate. The specific objectives are as follows:

1. Optimization and determination of the most suitable formulation for pineapple concentrate incorporating mature coconut water, utilizing scientific methods.
2. Evaluation of the sensory attributes (organoleptic properties) of the optimized formulation through rigorous sensory analysis techniques.
3. Assessment of the physico-chemical properties, including Total Soluble Solids (TSS), pH level, and microbiological profile (yeast and mold count) of the optimized pineapple concentrate infused with mature coconut water.
4. Investigation of consumer acceptability regarding the optimized formulation of pineapple concentrate with mature coconut water, employing validated consumer testing methodologies.

By addressing these objectives with scientific rigor, this study aims to provide empirical insights into the feasibility and desirability of utilizing mature coconut water in enhancing the quality and acceptability of pineapple concentrate.

(11) Review of Literature/Prior Art Search

Matured Coconut Water

Coconut (*Cocos nucifera* L.) is an important fruit tree found in tropical regions and its fruit can be made into a variety of foods and beverages. Normally people in tropical countries drink tender coconut water like a beverage as a sports drink. However, mature coconut water (MCW) is normally discharged as only the coconut meat is used for producing coconut milk. For instance, in Thailand MCW is a by-product from the coconut milk industry that might be used to produce nata decoco (Cappelletti et al., 2015).

According to (F. Ismail N. A. Et al., 2019). Matured coconut water is an agro waste that produced mainly from coconut milk industry which contains variety of sugars and minerals. This study was carried out on synthesizing levulinic acid from matured coconut water by focusing on the effect of catalytic acid to coconut water ratio, types of acid and reaction time.

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Coconut water may be an old commodity which is usually consumed as fresh, pasteurized coconut water (PCW) becomes a new entrant in the Philippine beverage industry. As a new player, penetrating the market of the giant beverage manufacturers seem to be risky and challenging. The study aimed to determine the market potential of bottled pasteurized coconut water and to identify product positioning opportunity in the beverage market. The study employed random consumption survey and product test. (Hanilyn A. Hidalgo 2017).

MCW: Health Benefits

May benefit to manage diabetes- Research shows that Coconut water may help to manage blood sugar levels and improve diabetes symptoms. It is a good source of manganese, which may increase insulin sensitivity and reduce blood sugar levels. **Help in good digestion-** Coconut water is rich in manganese that helps to regulate your bowel movements. It may prevent gaseous distension of the abdomen, constipation and acidity. **Reduce fatigue and stress-** There are instances such as fatigue, stress where you feel lethargic and don't want to eat or do anything. (Dr.Nikita Toshi 2023).

[CN103202505A](#) **Method for preparing recovery coconut water beverage**

The invention discloses a method for preparing recovery coconut water beverage. The method comprises the steps of: (1) extracting coconut water; (2) determining components; (3) standardizing the coconut water; (4) preparing the beverage; and (5) sterilizing the beverage according to the conventional film separation and ozone sterilization method, and performing conventional germ-free filling.

[CN107114628A](#) **Method for recycling old coconut water and coconut water syrup product**

The invention belongs to the technical field of food processing and discloses a method for recycling old coconut water and a coconut water syrup product. The method for recycling old coconut water comprises the following steps: cleaning surfaces of fresh or old coconuts, shelling to take water, and filtering to remove impurities in the coconut water; concentrating the filtered old coconut water in vacuum concentration or normal pressure concentration equipment, and heating;

2.3 Pineapple

Pineapple (*Ananas comosus*) is a large tropical fruit with a spiky, tough skin and sweet insides. When European explorers came across it in South America in the 17th century, they called it a pineapple

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because of its pinecone appearance. The scaly bumps on the outside are called "eyes." Slice it open, and you'll find bright yellow flesh that's both sweet and tart. For centuries, pineapples were so rare that only very wealthy people could afford to buy them. Some people even rented the exotic fruit. (Mikstas Christine, RD, LD 2022).

Pineapple, scientifically known as *Ananas Comosus* Merr., is one of the country's most popular fruits. It is also one of the top earners of foreign exchange among the country's agricultural commodities. The fruit contains water, carbohydrates, substantial amount of Vitamin C and potassium, and other nutrients. (agrimag 2019).

2.4 Pineapple: Health Benefits

May Reduce Blood Cholesterol. A preliminary review of several tropical fruits describes lower serum lipid profiles, triglycerides, and low-density lipoprotein (LDL) in rats who were given pineapple extract every day for 30 days. The authors also noted better cardiovascular health and lower inflammation as a result.

Decreased Inflammation. Like other types of fruit, pineapple is high in antioxidants, a feature that may reduce inflammation in the body (Cherney kristeen et al., 2022).

2.5 Concentrate. Is made from mostly fresh fruit. Fruit concentrate is produced from filtering the water out of fruit pulp. The concentrate product is stored under aseptic, non-aseptic and frozen conditions. These methods have been historically used to optimize its transportation across the world, to then be re-constituted with the same volume of water that was evaporated from it, once it reaches the location of juice manufacture. (Ogunsola Adeoti 2020).

2.6 Hot Filling. Hot-fill is a packaging and processing technique in which heat is used to sterilize the food product and subsequently the inner surface of the container during the filling process. It is a technique that is used extensively in the food packaging industry for a variety of applications ranging from applesauce to dips and sauces. The hot-fill process starts by initially heating the food product up to a temperature between 194-204°F as a method of killing bacteria and microorganisms present in the food (Moren Michael 2020).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

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The theoretical framework for the study on the acceptability of incorporating mature coconut water into making pineapple concentrate can draw upon several theoretical perspectives and concepts. Here are some potential elements of the theoretical framework:

1. Sensory Evaluation Theories: Drawing from sensory science, theories such as signal detection theory and psychophysics can provide a theoretical basis for understanding how sensory attributes influence consumer perception and acceptance of food products.
2. Acceptance and Preference Theories: The study may utilize theoretical frameworks such as the Expectancy-Confirmation Theory or the Theory of Planned Behavior to explore factors influencing consumer acceptance and preference for novel food products like pineapple concentrate with mature coconut water.
3. Food Product Development Theories: Theoretical perspectives from food science and technology, including principles of formulation optimization, sensory analysis, and quality control, can inform the experimental design and methodology for developing and evaluating the pineapple concentrate formulations.
4. Nutritional and Functional Food Theories: Theoretical frameworks related to the nutritional and functional properties of food, such as the concept of functional foods, can inform discussions around the potential health benefits and consumer perceptions of incorporating mature coconut water into pineapple concentrate.
5. Consumer Behavior Theories: Theoretical perspectives from consumer behavior research, such as the Theory of Consumer Behavior or the Consumer Decision-Making Process, can help elucidate the factors influencing consumer preferences, attitudes, and purchasing decisions related to novel food products.
6. Microbiological and Quality Assurance Theories: Theoretical frameworks related to food safety, microbiology, and quality assurance can guide discussions around microbiological analysis methods, shelf-life determination, and quality control measures for the pineapple concentrate formulations.

By integrating relevant theoretical perspectives from these domains, the study's theoretical framework can provide a comprehensive understanding of the factors influencing the acceptability,

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quality, and consumer perception of pineapple concentrate incorporating mature coconut water. This theoretical foundation will inform the study's research questions, hypotheses, experimental design, and interpretation of findings.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and

Procedure

The following were the steps followed in the preparation of the mature coconut water:

Preparation of the matured coconut water:

1. Collect coconut water and filter it.
2. Boil for 10 minutes and set aside.

Preparation of pineapple:

1. Peel the pineapple and remove the pineapple surface.
2. Cut the pineapple into cubes.
3. Put the pineapple in a blender and blend until pureed.
4. Set aside.

Preparation of seaweed extract:

Below are the steps followed in the preparation of seaweed extract:

1. Wash seaweed thoroughly with running water five times to remove adherent dirt and debris.
2. Drain for 5 minutes and weigh.
3. The seaweeds will be soaked in 6 cups of boiling water together with 1 cup of calamansi extract for about 1 hour and 30 minutes.
4. Strain to separate the seaweed extract.

Preparation of matured coconut water and pineapple juice concentrate with seaweed extract.

1. In a casserole, put the coconut water and sugar, then boil for 10 minutes.
2. Add the pineapple puree and seaweed extract and bring to a boil for about 15 minutes.
3. Remove the concentrate from the casserole and hot-fill it into the glass bottle.

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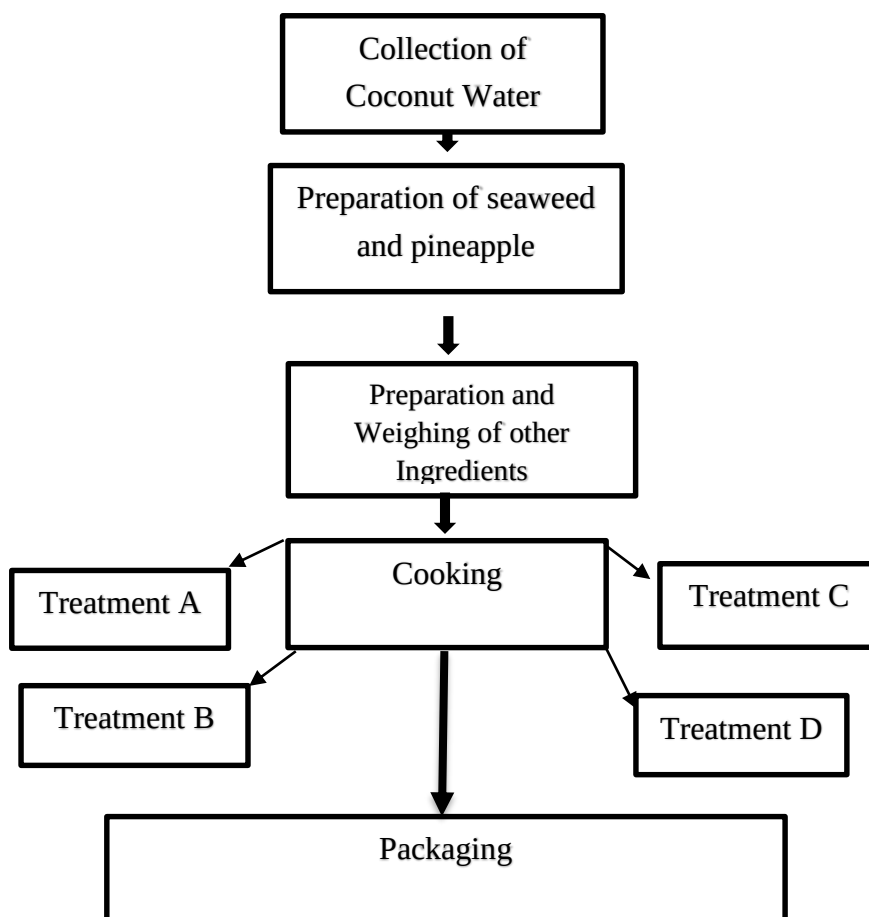


Figure 1. Flow Chart in Acceptability of Matured Coconut Water in Making Pineapple Juice Concentrate

Experimental Method and Design

In this study, we employ a systematic approach integrating experimental methods, design, and statistical analysis to comprehensively evaluate the acceptability of incorporating mature coconut water into the production of pineapple concentrate. Our investigation encompasses multiple facets, each addressed through tailored methodologies to ensure scientific rigor and robustness of findings.

1. Optimization of Formulation:

- Experimental Method: Design of Experiments (DOE)

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- Experimental Design: Factorial Design or Response Surface Methodology
- Statistical Analysis: Analysis of Variance (ANOVA), Response Surface Methodology (RSM), Regression Analysis
- Description: By systematically varying the proportions of pineapple juice and mature coconut water using DOE, we aim to identify the optimal formulation for pineapple concentrate. Statistical analyses, including ANOVA and RSM, will enable us to determine significant factors and interactions influencing acceptability and develop predictive models for formulation optimization.

2. Evaluation of Organoleptic Properties:

- Experimental Method: Sensory Analysis
- Experimental Design: Randomized Complete Block Design (RCBD)
- Statistical Analysis: Descriptive Statistics, Analysis of Variance (ANOVA), Principal Component Analysis (PCA)
- Description: Through sensory evaluation employing RCBD, trained panelists will assess the sensory attributes of pineapple concentrate formulations. Statistical analyses, including descriptive statistics and ANOVA, will summarize and compare sensory data. PCA will elucidate underlying patterns in sensory attributes to discern key factors driving consumer preferences.

3. Assessment of Physico-Chemical Properties and Microbiological Profile:

- Experimental Method: Laboratory Analysis
- Experimental Design: Completely Randomized Design (CRD)
- Statistical Analysis: Descriptive Statistics, T-tests or ANOVA, Correlation Analysis
- Description: Laboratory analyses following CRD will determine physico-chemical properties and microbiological profiles of pineapple concentrate formulations. Statistical analyses, including descriptive statistics and hypothesis testing, will elucidate differences among formulations and explore relationships between parameters.

4. Investigation of Consumer Acceptability:

- Experimental Method: Consumer Testing
- Experimental Design: Randomized Controlled Trial (RCT)
- Statistical Analysis: Descriptive Statistics, Chi-square Test, ANOVA
- Description: Consumer acceptability of pineapple concentrate formulations will be assessed through RCT. Statistical analyses, including descriptive statistics, chi-square tests, and ANOVA,

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• Research Assistant	2RA	22,000	
• Coconut water	50 lit.	1000	
• Glacelaria Seaweed	1 sack	1000	
• Pineapple fruit	10 kg	2000	
• Sugar	20kg.	2000	
• Lemon/Calamansi	5 kg.	1000	
• Sensory Plastic Cups with cover	200 cups	1000	
• Packaging materials	Industrial packaging to sachet packaging	50,000.00	
• Physico chemical test	Ph/tss	1000	
• Microbial analysis	Yeast and mold count	3000	
• Nutrifacts		10,000	
• Transportation Allowance		2000	
• Ph meter		70,000	
Total		P166,000.00	

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(21) Capsule Curriculum Vitae (One-page brief C.V. only)

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